BIO202 MIDTERM SUBJECTIVE

Qno1. Write down characteristics of enantiomers.

Enantiomers have nearly identical chemical properties but differ in a characteristic physical property: – their interaction with planepolarized light
Thus the presence of asymmetric carbon atoms also confers optical activity on the compound
When a beam of plane-polarized light is passed through a solution of an optical isomer, it rotates either to the right, dextrorotatory (+), or to the left, levorotatory (–)

Qno2: Name two diseases caused by misfolding of proteins.

- Alzheimer's disease
- Type 2-diabetes,
- Huntington's diseas
- Parkinson's disease, arise from a common misfolding mechanism.

Qno3: What are digestive proteins? Explain with examples.

Dietary proteins are very large complex molecules that cannot be absorbed from the intestine. To be absorbed, dietary proteins must be digested to small simple molecules (amino acids), which are easily absorbed from the intestine. I-Digestion in the stomach, example

Qno4: What is ketogenic amino acid?

Amino acids whose catabolism yields either acetoacetate or its precursor, (acetyl CoA or acetoacetyl CoA) are termed ketogenic. Acetoacetate is one of the ketone bodies which also include 3-hydroxybutyrate and acetone.

Qno5: Give characteristics of furfural compounds.

1. The furfural products thus formed can condense with certain organic phenols (α-naphthol) to form compounds having characteristic colours. This forms basis of certain tests used for detection of sugars
2. Pentoses yield a cyclic aldehyde furfural with 12% HCL. Hexoses are decomposed by hot strong acids to give hydroxy methyl furfural

Qno6: Name factors that influence free energy of protein molecules.

The free energy of a protein molecule is influenced by
(1) The hydrophobicity,
(2) Hydrogen bonds,
(3) Electrostatic interactions
(4) The conformational entropy due to the restricted motion

Qno7: What do u knows about allosteric amino acid?

In Greek, Allo means other and Steric means Site. Apart from Oxygen, ability of hemoglobin to reversibly bind oxygen is affected by – the pH of the environment – the p CO_2; and – the availability of 2,3-

VU Medical Zone Effort By Amaan Khan

bisphosphophoglycerate. These are collectively called Allosteric (other site) effectors, because their interaction at one site on the hemoglobin molecule affects the binding of oxygen to heme groups at other locations on the molecule.

Qno8: Define allotropy?

The existence of a substance in more than one crystalline depending upon physical properties is called allotropy.

Qno10: What is HSP60?

The hsp60 family of chaperones, sometimes called chaperonins, differ in sequence and structure from hsp70 and its homology. Hsp60 acts later in the folding process, often together with an hsp70 chaperone. Hsp60 chaperonins form a multi-subunit barrel-shaped structure. The unfolded protein fits into the barrel cavity that excludes water and serves as a template for the folding process.

Qno11: How Duchenne Muscular Dystrophy is caused?

Duchenne Muscular dystrophy results from a large segment of gene deletion in X chromosome, resulting in small truncated dystrophin protein. The defect can range from a single change in the amino acid sequence (as in sickle cell anemia) to deletion of a larger portion of the polypeptide chain (as in most cases of Duchenne muscular dystrophy)

Qno12: Difference between Hemiacetals and Hemiketals?

The reaction can produce either of two stereoisomeric configurations, denoted α and β. Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers, and the carbonyl carbon atom is called the anomeric carbon.

Qno13: How misfolding of proteins occurs?

Misfolding of proteins may occur spontaneously or caused by a mutation in a particular gene, producing an altered protein. In addition some apparently normal proteins can, after abnormal proteolytic cleavage, take on a unique conformational state and that leads to the formation of long fibrillar protein assemblies consisting of β pleated sheets.

Qno14: What are monosaccharides? Write two examples.

Monosaccharides (simple sugars) are those which cannot be hydrolyzed further into simpler forms. The backbones of common monosaccharides are unbranched carbon chains in which all the carbon atoms are linked by single bonds. Example Glucose and Fructose

Qno15: What are Chaperons and write its main function.

As the protein folds and refolds while it is searching for its native low energy state, it passes through many highenergy conformations that slow the process. These high energy conformations are called kinetic barriers. Cells employ auxiliary proteins to overcome these kinetic barriers. These auxillary proteins are called chaperones.

Function: Chaperone use energy provided by ATP hydrolysis to assist in the folding process. Chaperones participate in the folding of over half of mammalian proteins. They prevent immature folding of the nascent polypeptide and help in the final native protein formation.

EFFORT BY AMAAN KHAN

W Medical Zone Effort By Amaan Khan

Q What are carbohydrates? and write four major groups of carbohydrates.

Qno16: What is anomers and anomeric carbon?

Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers. The carbonyl carbon atom is called the anomeric carbon. Carbon 1, after cyclization has four different groups attached to it and thus it also becomes 'asymmetric' Carbohydrates • The newly created assymetric or chiral carbon is called the anomeric carbon.

Qno17: What are starch and its structure?

Starch contains two types of glucose polymers: – Amylose – amylopectin. It is the most important dietary carbohydrate in cereals, potatoes, legumes, and other vegetables. Amylose (13–20%), consists of long, unbranched helical structure of D-glucose residues connected by (α1→4) linkages (as in maltose).

Qno18: Explain beta bends.

Beta-bends are short regions usually involving four successive amino acid residues. They often connect strands of antiparallel betasheets. β-Bends reverse the direction of a polypeptide chain, helping it form a compact, globular shape.

Qno19: Explain side chain importance of alanine.

Amino acids most commonly found in alpha helices are non-polar with aliphatic side chains. Alanine shows the greatest tendency to form α helices in most experimental model systems.

Q20: Define domain in protein.

A domain is a section of protein structure sufficient to perform a particular chemical or physical task such as binding of a substrate or other ligand.

Q21: Explain protein structure.

Proteins are big molecules. The covalent backbone of a typical protein contains hundreds of individual bonds. Because free rotation is possible around many of these bonds, the protein can assume an unlimited number of conformations. However, each protein has a specific chemical or structural function, strongly suggesting that each has a unique three-dimensional structure. The spatial arrangement of atoms in a protein is called its conformation. The possible conformations of a protein include any structural state that can be achieved without breaking covalent bonds. A change in conformation could occur, for example, by rotation about single bonds. Proteins in any of their functional, folded conformations are called native proteins. In the context of protein structure, the term stability can be defined as the tendency to maintain a native conformation. The chemical interactions that counteract these effects and stabilize the native conformation include disulfide bonds and the weak (noncovalent) interactions hydrogen bonds, and hydrophobic and ionic interactions. About 200 to 460 kJ/mol are required to break a single covalent bond, whereas weak interactions can be disrupted by a mere 4 to 30 kJ/mol. Individual covalent bonds that contribute to the native conformations of proteins, such as disulfide bonds linking separate parts of a single polypeptide chain, are clearly much stronger than individual weak interactions. Yet, because they are so numerous, it is weak interactions that predominate as a stabilizing force in protein structure. On carefully examining the contribution of weak interactions to protein stability, we find that hydrophobic interactions generally predominate. Hydrophobic amino acid side chains therefore tend to cluster in a protein's interior, away from water, forming a hydrophobic protein core. Hydrogen bonds between groups in a protein form cooperatively

VU Medical Zone Effort By Amaan Khan

(formation of one makes the next one more likely) in repeating secondary structures that optimize hydrogen bonding The interaction of oppositely charged groups that form an ion pair, or salt bridge, can have either a stabilizing or destabilizing effect on protein structure. Ionic interaction also limit structural flexibility and confer a uniqueness to a particular protein structure. In the tightly packed atomic environment of a protein, one more type of weak interaction van der Waals interactions. Van der Waals interactions are dipoledipole interactions involving the permanent electric dipoles in groups such as carbonyls.–can have a significant effect The six atoms of the peptide group lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other. The peptide CN bonds are unable to rotate freely. The rigid peptide bonds limit the range of conformations that can be assumed by a polypeptide chain.

Qno22: What are carbohydrates? Name their major groups.

Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis.
Carbohydrates are classified into four major groups:
Monosaccharides
Oligosaccharides
Disaccharides
Polysaccharides

Qno23: Explain basics of classification of proteins.

• Different classifications of proteins are based on their:
1. Shape and Size
2. Biological actions/ Functions
3. Solubility and physical properties
4. Quality
1. Catalytic Proteins: These specialized proteins are called enzymes which catalyze the biochemical reactions
2. Protective Proteins a) Immunoglobins (Igs) • These freely circulating proteins protect the body from invading microbes such as bacteria or viruses by inactivating or killing them through various mechanisms.
b) Fibrinogen • This forms fibrin clot and stops bleeding from wounds
3. Regulatory Proteins: • Hormones control genetic expression, cellular signalling and biochemical reactions catalyzed by enzymes. Enzymes are either activated or inactivated through modification of their structure.
Such as Insulin Glucagon Somatostatin
Structural Proteins: These proteins form various body structures e.g. Collagen Elastin Keratin
5. Transport Proteins: These proteins transport various substances from one part of the body to the other e.g. Hemoglobin transport O2from lungs to tissues and CO2 from tissues to lungs Transferrin transports iron
6. Contractile Proteins: These proteins are involved in muscle contraction and relaxation − Myosin of thick filaments − Actin of thin filaments of skeletal muscles.

VU Medical Zone Effort By Amaan Khan

Qno24: What are diasaccharide two examples. 3

Disaccharides are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis, such as Maltose and Sucrose.

Q 25: Explain quatnary structute of protein.

Some proteins contain two or more separate polypeptide chains, or subunits, which may be identical or different. The arrangement of these protein subunits in three-dimensional complexes constitutes quaternary structure. In considering these higher levels of structure, it is useful to classify proteins into two major groups:

fibrous proteins, having polypeptide chains arranged in long strands or sheets, 24 and globular proteins, having polypeptide chains folded into a spherical or globular shape. The two groups are structurally distinct: fibrous proteins usually consist largely of a single type of secondary structure; globular proteins often contain several types of secondary structure. The two groups differ functionally in that the structures that provide support, shape, and external protection to vertebrates are made of fibrous proteins, whereas most enzymes and regulatory proteins are globular proteins. Fibrous proteins share properties that give strength and/or flexibility to the structures in which they occur. All fibrous proteins are insoluble in water, a property conferred by a high concentration of hydrophobic amino acid residues both in the interior of the protein and on its surface. The -keratins have evolved for strength. Found in mammals, these proteins constitute almost the entire dry weight of hair, wool, nails, claws, quills, horns, hooves, and much of the outer layer of skin. α-keratin is rich in the hydrophobic residues Ala, Val, Leu, Ile, Met, and Phe. In α-keratins, the cross-links stabilizing quaternary structure are disulfide bonds.

Qno26: How alpha helix stablize.tautomers?

The right-handed α helix is the common form. Extended left-handed α helices are theoretically less stable and have not been observed in proteins. The α helix proved to be the predominant structure in α-keratins. The structure is stabilized by a hydrogen bond between the hydrogen atom attached to the electronegative nitrogen atom of a peptide linkage and the electronegative carbonyl oxygen atom of the fourth amino acid on the amino-terminal side of that peptide bond.

Q27: Explain the biological importance of carbohydrates.

Biomedical Importance
• Chief source of energy
• Constituents of compound lipids and conjugated proteins
• Lactose, the principal sugar of milk
• Degradation products utilized for synthesis of other substances such as fatty acids, amino acids, cholesterol, etc.
• Constituents of mucopolysaccharide s which form the ground substance of mesenchymal tissues
• Certain carbohydrate derivatives are used as drugs like cardiac glycosides.
• Inherited deficiency of certain enzymes in metabolic pathways of different carbohydrates can cause diseases e.g. galactosemia, glycogen storage diseases, lactose intolerance, etc
• Derangement of glucose metabolism is seen in Diabetes mellitus.

Qno28: What is role proton bind with heme group in tissues and lung?

Proton Binding:
• An increase in protons and\or a higher pCO_2 shifts the equilibrium to the right
• Therefore favoring deoxyhemoglobin T form.
• Whereas a decrease in pCO_2 and\or a decrease in protons shifts the equilibrium to the left
• Conversely, an increase in p O_2 promotes proton release
• The differential pH gradient (lungs having a higher pH, tissues a lower pH) favors the unloading of oxygen in the peripheral tissues,
• And the loading of oxygen in the lung
• Binding of oxygen to facilitate other molecule of oxygen is called positive cooperativity
• Whereas, binding of carbon dioxide and proton to release oxygen is called negative allostery, in which;
• Increased CO_2 and H+ ion favor oxygen unloading.
• Thus we see that the four polypeptide chains of hemoglobin communicate with each other about not only O_2 binding to their heme groups

VU Medical Zone Effort By Amaan Khan